Coordinate-Wise Additivity and the ℓ^1 Norm on Finite Graph Cochains

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Abstract

We prove that on finite directed graphs, any system of seminorms on the cochain space $C^1(G, \mathbb{R}^k) = \bigoplus_{e \in E} \mathbb{R}^k$ satisfying edge locality, coordinate permutation symmetry, and disjoint-coordinate additivity is coordinatewise ℓ^1 on each edge. Across edges the general conclusion is orbit-weighted ℓ^1; a single global scalar requires edge-transitivity or an explicit uniformity/naturality assumption across edge types. The additivity assumption is justified only for coordinate-separable, replica-extensive observers. Relaxing those assumptions yields alternative obstruction geometries.

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts; scalar absolute values are denoted $|\cdot|$.

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1 Introduction

This paper is the first step in a sequence analyzing projection-induced defects on presheaf coboundaries through progressively richer structural settings:

- **This paper** establishes that coordinate-wise additivity and symmetry on finite graph cochains select coordinatewise ℓ^1 geometry within the coordinate-separable observer class. The result is purely combinatorial and requires no functional-analytic or dynamical structure.
- **Paper 001** [7] extends this classification to Banach presheaves, showing that functoriality under bounded linear maps similarly forces the ℓ^1 coboundary norm as the unique additive diagnostic compatible with the ambient Banach geometry.
- **Paper 000** [6] applies this rigidity to dynamics, proving that projection of linear evolution onto a nonlinear manifold produces intrinsic coboundary defects whose natural measurement is therefore necessarily ℓ^1.

Although numbered `004` in the repository, this paper is logically prior to the Banach-presheaf and projection-obstruction papers. The numbering reflects repository order, not dependency order.

The purpose of the present paper is to isolate the minimal combinatorial realization of the replica-extensive, coordinate-separable observer class behind this rigidity. In the broader M-compatible framework, this corresponds to the case where coordinate channels are required to aggregate independently. Other symmetry classes yield different admissible gauges. Rather than

beginning with Banach space assumptions or dynamical systems, we show that ℓ^1 geometry is selected for finite graph cochains under coordinate-separable, replica-extensive evaluation.

A central modeling vulnerability is treating the summation of independent measurement channels as an ad hoc assumption (L^2 root-sum-squares assume an entirely different aggregation). We isolate that assumption rather than hiding it. When independent observations (the disjoint coordinate supports) are required to compose by the Disjoint Observer Consistency boundary, Cauchy's functional logic selects linear summation. Without that observer boundary, the conclusion does not follow.

2 ℓ^1 Characterization on Graph Cochains

We recall that a **seminorm** on a vector space V is a functional $\nu : V \to \mathbb{R}_{\geq 0}$ that is subadditive ($\nu(x+y) \leq \nu(x) + \nu(y)$) and absolutely homogeneous ($\nu(cx) = |c|\nu(x)$ for all scalars $c \in \mathbb{R}$).

Theorem 2.1. *For any finite directed graph $G = (V, E)$, consider the cochain space*

$$C^1(G, \mathbb{R}^k) = \bigoplus_{e \in E} \mathbb{R}^k. \tag{1}$$

Let $G \mapsto \nu_G$ be an assignment of seminorms to finite directed graphs, invariant under graph isomorphism in the sense stated below, with

$$\nu_G : C^1(G, \mathbb{R}^k) \to \mathbb{R}_{\geq 0}, \tag{2}$$

satisfying:

1. ***Edge locality (local additivity)***

$$\nu_G(\delta) = \sum_{e \in E} \nu_e(\delta_e) \tag{3}$$

2. ***Coordinate symmetry***

$$\nu_e(x_{\sigma(1)}, \ldots, x_{\sigma(k)}) = \nu_e(x_1, \ldots, x_k) \tag{4}$$

   for every permutation σ.

3. ***Coordinate additivity (observer assumption justified by Disjoint Observer Consistency)***
   If an observer combines $x, y \in \mathbb{R}^k$ which have entirely disjoint coordinate support ($\mathrm{supp}(x) \cap \mathrm{supp}(y) = \varnothing$), and if that observer treats those supports as independently additive, then:

$$\nu_e(x + y) = \nu_e(x) + \nu_e(y). \tag{5}$$

4. ***Graph invariance***
   For any graph isomorphism $\sigma : G \to G'$,

$$\nu_{G'}(\sigma_* \delta) = \nu_G(\delta). \tag{6}$$

Then there are positive weights $w_{\mathcal{O}}$, one for each edge orbit $\mathcal{O}$ under graph isomorphisms internal to the admissible graph class, such that

$$\nu_G(\delta) = \sum_{\mathcal{O}} w_{\mathcal{O}} \sum_{e \in \mathcal{O}} \|\delta_e\|_1. \tag{7}$$

If the graph/action is edge-transitive, or if a global edge-uniformity axiom is imposed across all edge types, this reduces to a single scalar

$$\nu_G(\delta) = \alpha \sum_{e \in E} \|\delta_e\|_1. \tag{8}$$

Remark. Axiom 3 guarantees the system tracks as an abstract L-space [1]. Standard models often insert this assumption implicitly; here it is isolated as an explicit observer condition. When evaluations are required to aggregate independently across boundary partitions without cross-domain coupling, the additivity conclusion follows.

Proof. By edge locality it suffices to classify the seminorm ν_e on $\mathbb{R}^k$. Let $x = (x_1, \ldots, x_k)$. Decompose

$$x = \sum_{i=1}^{k} x_i \hat{e}_i. \tag{9}$$

The vectors $x_i \hat{e}_i$ have disjoint coordinate support, so coordinate additivity gives

$$\nu_e(x) = \sum_{i=1}^{k} \nu_e(x_i \hat{e}_i). \tag{10}$$

Since ν_e is a seminorm, it is absolutely homogeneous, providing

$$\nu_e(x_i \hat{e}_i) = |x_i| \nu_e(\hat{e}_i). \tag{11}$$

By coordinate symmetry, $\nu_e(\hat{e}_i) = w_e$ for all i. Hence

$$\nu_e(x) = w_e \sum_{i=1}^{k} |x_i| = w_e \|x\|_1. \tag{12}$$

Thus across the entire graph,

$$\nu_G(\delta) = \sum_{e \in E} w_e \|\delta_e\|_1. \tag{13}$$

Finally, graph invariance implies that weights are constant on edge orbits. Graph isomorphism invariance alone does not compare non-equivalent edge orbits, and it does not compare unrelated graph families unless the seminorm assignment is required to be natural under the chosen embeddings/disjoint-union completions. Thus the general conclusion is

$$\nu_G(\delta) = \sum_{\mathcal{O}} w_{\mathcal{O}} \sum_{e \in \mathcal{O}} \|\delta_e\|_1. \tag{14}$$

Under edge-transitivity, or under an explicit global normalization identifying all edge types, $w_{\mathcal{O}} = \alpha$ and the unweighted formula follows. □

Corollary 2.2 (Discrete Weighted L^1 Structure of the Defect Space)**.** *Under the hypotheses of Theorem 2.1, the diagnostic space is a finite weighted L^1 space. In the edge-uniform case it is proportional to*

$$(C^1(G, \mathbb{R}^k), \nu_G) \cong L^1(E \times \{1, \ldots, k\}) \tag{15}$$

with respect to the counting measure.

Proof. By Theorem 2.1 in the edge-uniform case,

$$\nu_G(\delta) = \alpha \sum_{e \in E} \|\delta_e\|_1. \tag{16}$$

Writing $\delta_{e,i}$ for the i-th coordinate of δ_e, we obtain

$$\nu_G(\delta) = \alpha \sum_{e \in E} \sum_{i=1}^{k} |\delta_{e,i}|. \tag{17}$$

Define $f : E \times \{1, \ldots, k\} \to \mathbb{R}$ by $f(e, i) = \delta_{e,i}$. Then

$$\nu_G(\delta) = \alpha \|f\|_{L^1}. \tag{18}$$

Thus the diagnostic space is isometrically equivalent to $L^1(E \times \{1, \ldots, k\})$ under the counting measure in the uniform case. In the general orbit-weighted case it is a weighted finite L^1 space. □

3 Structural Consequences

3.1 Coarea Duality ($k = 1$)

For scalar signals, the classified norm has a distinguished topological property.

Fact (Discrete Coarea Formula [2, 3])**.** For $v : V \to \mathbb{R}$:

$$\mathrm{TV}_G(v) = \int_{-\infty}^{\infty} |\partial V_t| \, dt \tag{19}$$

where $V_t = \{m : v_m > t\}$ and $|\partial V_t|$ counts edges crossing the boundary.

When boundary conditions force a binary partition ($v_a = 1, v_b = 0$), ℓ^1 minimization recovers the minimum s-t cut via max-flow/min-cut [5].

Remark. For $k > 1$, the total variation $\sum_e \|v_n - v_m\|_1$ decomposes (by Theorem 2.1) as a sum of k independent scalar total variations, each individually satisfying the coarea formula.

3.2 The Geometry of the Defect Field

The explicit inclusion of **Coordinate Additivity** formally captures the declared independence of distinct observed coordinates. Permutation symmetry and absolute homogeneity alone permit alternative norms such as ℓ^∞ or ℓ^2. The strict structural requirement that isolated defect channels accumulate their structural signatures linearly thereby determines the ℓ^1 stalk geometry under these constraints.

Thus, the seminorm compatible with the stated local coordinate-additivity and coordinate-symmetry constraints is coordinatewise ℓ^1 on each edge. Uniform ℓ^1 over the product index set requires the additional edge-uniformity assumption stated in Theorem 2.1.

Conceptually, our theorem establishes (cf. [4]) that the defect space restricts to:

$$C^1(G, \mathbb{R}^k) \cong L^1(E \times \{1, \ldots, k\}) \tag{20}$$

whenever the diagnostic respects independent components and edge weights are uniform. More generally the result is a finite weighted L^1 geometry. This serves as a finite combinatorial analogue of Kakutani's (1941) concrete characterization of abstract L-spaces [1]. Local additivity across independent structural components selects ℓ^1-type geometry over the relevant finite measure. Defect fields in the uniform scalar case behave as discrete 1-forms whose absolute total variation tracks graph cuts and sparse obstructions [3].

3.3 Role in the Broader Classification Program

The result obtained here isolates the combinatorial origin of ℓ^1-type geometry as a consequence of coordinate-wise and edge-wise separability. In subsequent work, this principle is generalized. In particular, when the cochain spaces are replaced by Banach presheaves and coordinate additivity is replaced by structural compatibility in the form of functoriality under bounded linear maps [7], the same weighted ℓ^1 edge-sum structure is recovered, with uniformity requiring additional symmetry or normalization.

The present theorem should not be interpreted as asserting universality of ℓ^1 across all observers. Rather, it identifies the unique geometry compatible with coordinate-wise independence and replica-extensive aggregation. In the general M-compatible classification, relaxing coordinate additivity admits alternative geometries such as ℓ^2 and mixed norms. The present theorem therefore provides the minimal combinatorial base case from which the more general Banach [7] and dynamical [6] results can be understood as specific structural extensions.

3.4 Conditional Characterization of ℓ^1

A compressed form of the result is the following. If a diagnostic Φ is coordinate-separable, refinement-invariant, homogeneous, and additive across independent coordinates, then it has the form

$$\Phi(x) = \sum_i \psi(x_i) \tag{21}$$

with ψ additive and regular. Hence $\psi(t) = C|t|$ and

$$\Phi(x) = C \sum_i |x_i|. \tag{22}$$

Thus ℓ^1 is selected by coordinate separability, additive aggregation, and regularity. Relaxing any of these assumptions admits alternative geometries.

4 Zenodo Description

This preprint proves a finite combinatorial classification theorem for obstruction diagnostics on graph cochains. For a finite directed graph G and cochain space $C^1(G, \mathbb{R}^k)$, any seminorm satisfying edge locality, coordinate permutation symmetry, and additivity over disjoint coordinate supports is coordinatewise ℓ^1 on each edge. Across edges, the general conclusion is orbit-weighted ℓ^1; a single global scalar requires edge-transitivity or an explicit edge-uniformity assumption.

The result isolates the minimal coordinate-separable, replica-extensive observer assumptions under which ℓ^1 geometry is selected. It does not claim that ℓ^1 is universal. Relaxing coordinate additivity, edge locality, symmetry, or uniformity admits alternative geometries such as ℓ^2, ℓ^∞, mixed norms, or weighted variants.

This paper supplies the finite combinatorial base case for later obstruction-calculus papers on Banach presheaf coboundaries, M-compatible gauge classification, and projection-induced defects in factorized-state dynamics.

Keywords: finite graph cochains; coordinate-wise additivity; ℓ^1 norm; weighted ℓ^1; graph cohomology; finite directed graphs; seminorm classification; replica extensivity; coordinate-separable observers; edge locality; obstruction diagnostics; finite obstruction calculus; cohomological obstruction theory; total variation; discrete coarea formula; max-flow min-cut; M-compatible gauges.

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Changelog

v6.0 (May 2026): Added publication status box. Clarified dependency order and repository numbering. Softened over-strong physical/universal language. Clarified orbit-weighted versus edge-uniform L^1 in Corollary 2.2. Removed duplicated §3.2 paragraph. Polished the compressed conditional characterization in §3.4.
v5.0 (May 2026): Added cross-reference to Paper 000 v004 [8] (observer theory, contextual congruence, reflective completion). The coordinate-wise additivity result provides the combinatorial base case for the gauge classification hierarchy. Observer passivity (O1 in [8]) constrains downstream processing of the ℓ^1 diagnostic derived here.
v4.0 (April 2026): Audit-hardened pre-print. Clarified orbit-weighted vs. uniform ℓ^1.